Problem	Given Pressure	*Patm*	Express Result As:
3.14	583 kPa(abs)	103 kPa(abs)	Gage pressure
3.15	157 kPa(abs)	101 kPa(abs)	Gage pressure
3.16	30 kPa(abs)	100 kPa(abs)	Gage pressure
3.17	74 kPa(abs)	97 kPa(abs)	Gage pressure
3.18	101 kPa(abs)	104 kPa(abs)	Gage pressure
3.19	284 kPa(gage)	100 kPa(abs)	Absolute pressure
3.20	128 kPa(gage)	98.0 kPa(abs)	Absolute pressure
3.21	4.1 kPa(gage)	101.3 kPa(abs)	Absolute pressure
3.22	−29.6 kPa (gage)	101.3 kPa(abs)	Absolute pressure
3.23	−86.0 kPa (gage)	99.0 kPa(abs)	Absolute pressure
3.24	84.5 psia	14.9 psia	Gage pressure
3.25	22.8 psia	14.7 psia	Gage pressure
3.26	4.3 psia	14.6 psia	Gage pressure
3.27	10.8 psia	14.0 psia	Gage pressure
3.28	14.7 psia	15.1 psia	Gage pressure
3.29	41.2 psig	14.5 psia	Absolute pressure
3.30	18.5 psig	14.2 psia	Absolute pressure
3.31	0.6 psig	14.7 psia	Absolute pressure
3.32	−4.3 psig	14.7 psia	Absolute pressure
3.33	−12.5 psig	14.4 psia	Absolute pressure

Pressure–Elevation Relationship

3.34 If milk has a specific gravity of 1.08, what is the pressure at the bottom of a milk can 550 mm deep?

3.35 The pressure in an unknown fluid at a depth of 4.0 ft is measured to be 1.820 psig. Compute the specific gravity of the fluid.

3.36 The pressure at the bottom of a tank of propyl alcohol at 25° C must be maintained at 52.75 kPa(gage). What depth of alcohol should be maintained?

3.37 When you dive to a depth of 12.50 ft in seawater, what is the pressure?

3.38 A water storage tank is on the roof of a factory building and the surface of the water is 50.0 ft above the floor of the factory. If a pipe connects the storage tank to the floor level and the pipe is full of static water, what is the pressure in the pipe at floor level?

3.39 An open tank contains ethylene glycol at 25°C. Compute the pressure at a depth of 3.0 m.

3.40 For the tank of ethylene glycol described in Problem 3.39, compute the pressure at a depth of 12.0 m.

3.41 Figure 3.19 shows a diagram of the hydraulic system for a vehicle lift. An air compressor maintains pressure above the oil in the reservoir. What must the air pressure be if the pressure at point A must be at least 180 psig?

3.42 Figure 3.20 shows a clothes washing machine. The pump draws fluid from the tub and delivers it to the disposal sink. Compute the pressure at the inlet to the pump when the water is static (no flow). The soapy water solution has a specific gravity of 1.15.

3.43 An airplane is flying at 10.6 km altitude. In its nonpressurized cargo bay is a container of mercury 325 mm deep.

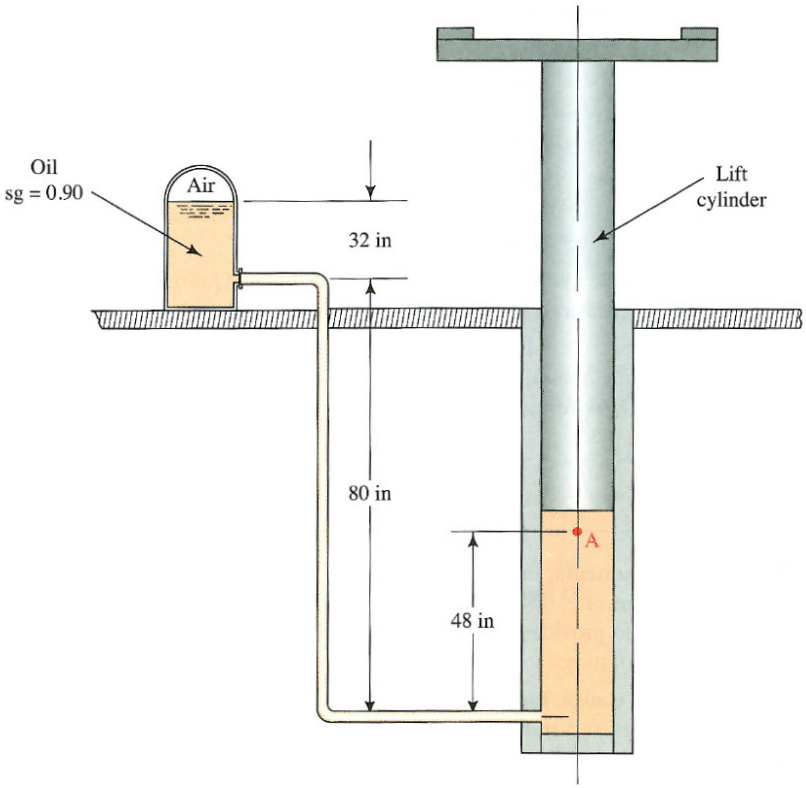


FIGURE 3.19 Vehicle lift for Problem 3.41.

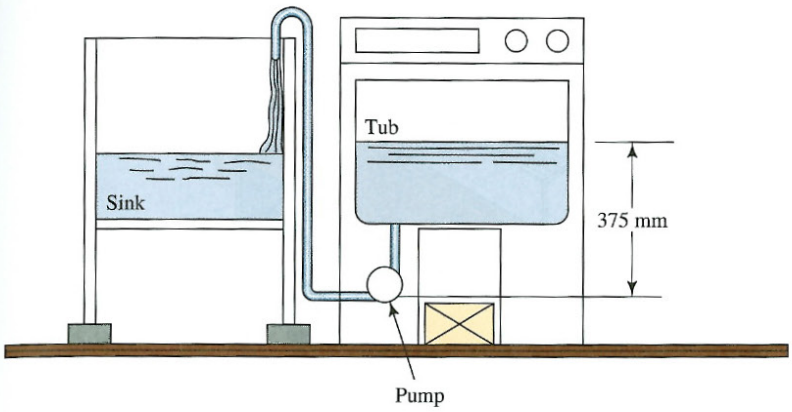


FIGURE 3.20 Washing machine for Problem 3.42.

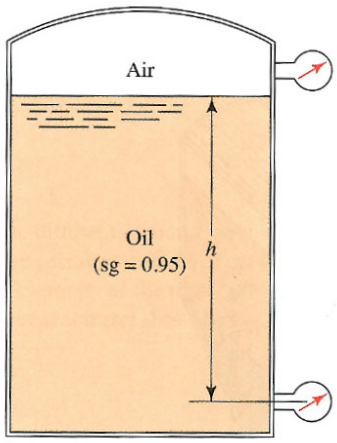


FIGURE 3.21 Problems 3.44–3.47.

The container is vented to the local atmosphere. What is the absolute pressure at the surface of the mercury and at the bottom of the container? Assume the conditions of the standard atmosphere prevail for pressure. Use sg = 13.54 for the mercury.

3.44 For the tank shown in Fig. 3.21, determine the reading of the bottom pressure gage in psig if the top of the tank is vented to the atmosphere and the depth of the oil h is 28.50 ft.

3.45 For the tank shown in Fig. 3.21, determine the reading of the bottom pressure gage in psig if the top of the tank is sealed, the top gage reads 50.0 psig, and the depth of the oil h is 28.50 ft.

3.46 For the tank shown in Fig. 3.21, determine the reading of the bottom pressure gage in psig if the top of the tank is sealed, the top gage reads −10.8 psig, and the depth of the oil h is 6.25 ft.

3.47 For the tank shown in Fig. 3.21, determine the depth of the oil h if the reading of the bottom pressure gage is 35.5 psig, the top of the tank is sealed, and the top gage reads 30.0 psig.

3.48 For the tank in Fig. 3.22, compute the depth of the oil if the depth of the water is 2.80 m and the gage at the bottom of the tank reads 52.3 kPa(gage).

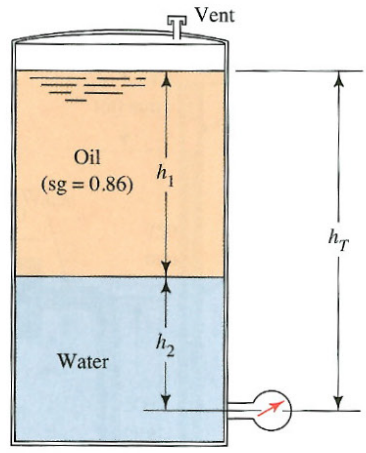


FIGURE 3.22 Problems 3.48–3.50.

3.49 For the tank in Fig. 3.22, compute the depth of the water if the depth of the oil is 6.90 m and the gage at the bottom of the tank reads 125.3 kPa(gage).

3.50 Figure 3.22 represents an oil storage drum that is open to the atmosphere at the top. Some water was accidentally pumped into the tank and settled to the bottom as shown in the figure. Calculate the depth of the water h_2 if the pressure gage at the bottom reads 158 kPa(gage). The total depth h_T = 18.0 m.

3.51 A storage tank for sulfuric acid is 1.5 m in diameter and 4.0 m high. If the acid has a specific gravity of 1.80, calculate the pressure at the bottom of the tank. The tank is open to the atmosphere at the top.

3.52 A storage drum for crude oil (sg = 0.89) is 32 ft deep and open at the top. Calculate the pressure at the bottom.

3.53 The greatest known depth in the ocean is approximately 11.0 km. Assuming that the specific weight of the water is constant at 10.0 kN/m^3, calculate the pressure at this depth.

3.54 Figure 3.23 shows a closed tank that contains gasoline floating on water. Calculate the air pressure above the gasoline.

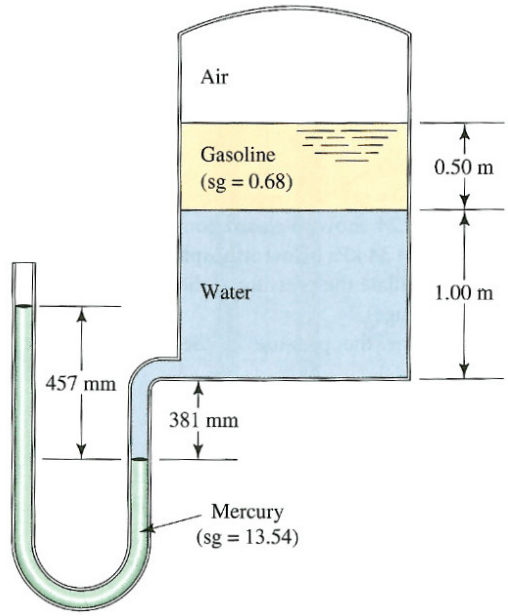


FIGURE 3.23 Problem 3.54.

FIGURE 3.24 Problem 3.55.

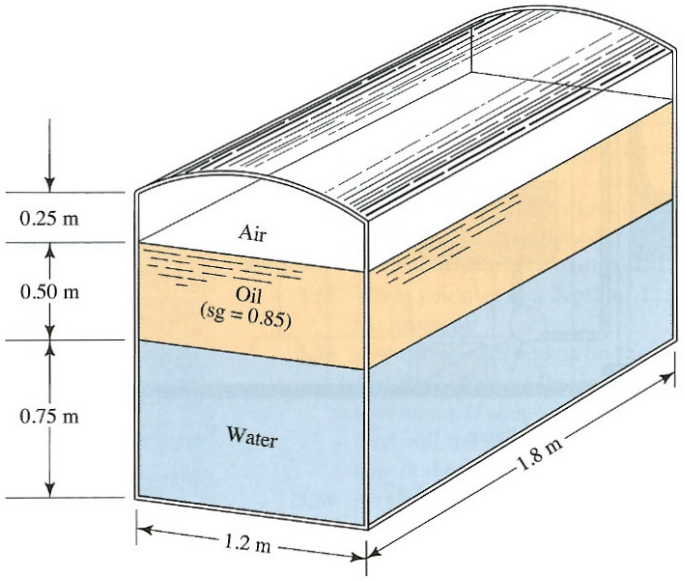
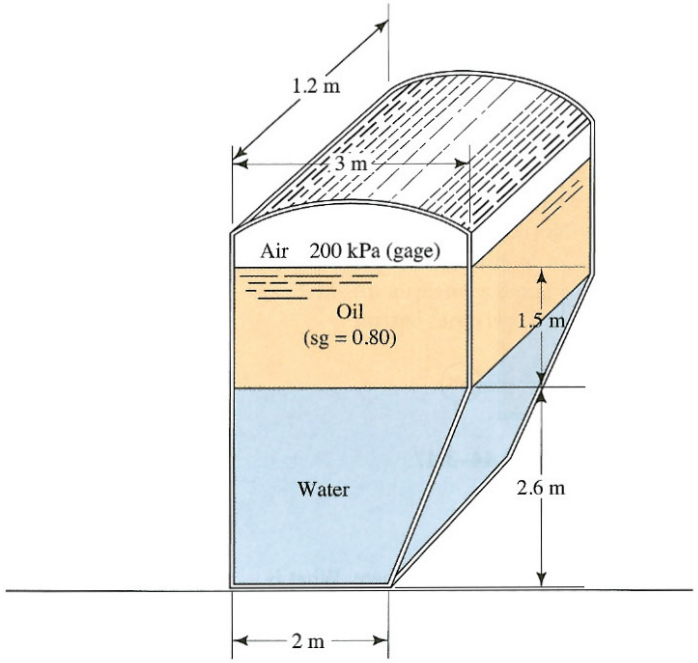


FIGURE 3.25 Problem 3.56.



3.55 Figure 3.24 shows a closed container holding water and oil. Air at 34 kPa below atmospheric pressure is above the oil. Calculate the pressure at the bottom of the container in kPa(gage).

3.56 Determine the pressure at the bottom of the tank in Fig. 3.25.

Manometers

3.57 Describe a simple U-tube manometer.

3.58 Describe a differential U-tube manometer.

3.59 Describe a well-type manometer.

3.60 Describe an inclined well-type manometer.

3.61 Describe a compound manometer.

3.62 Water is in the pipe shown in Fig. 3.26. Calculate the pressure at point A in kPa(gage).

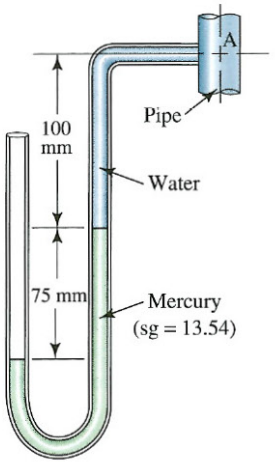


FIGURE 3.26 Problem 3.62.